

# Develop investigable questions, reasoned predictions and hypotheses to explore scientific models, identify patterns and test relationships

## Learning intention

- develop investigable questions, reasoned predictions and hypotheses to explore scientific models, identify patterns and test relationships.

## Teach and model

- predicting what will happen when conditions change in a given scenario or phenomenon.
- discussing what is meant by a causal relationship and examining how causation is different from correlation.
- formulating hypotheses such as: 'An earthquake of greater magnitude will cause more damage because there is more energy transferred'.

## Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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## Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

## Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.